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# Discovering Cooperative Pipelines: Autoresearch for Sequential Social Dilemmas

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## Abstract

1 We study *two-level autoresearch* for cooperation: an outer-loop AI agent au-  
2 tonomously redesigns the inner-loop pipeline of an LLM policy-synthesis sys-  
3 tem for multi-agent Sequential Social Dilemmas (SSDs). A *researcher agent*  $\mathcal{R}$   
4 (run as a coding agent) reads the inner-loop source code, edits system prompts,  
5 feedback functions, helper libraries, and iteration logic, runs evaluations, and  
6 decides what to keep, following the *autoresearch* paradigm. Across two games  
7 (Cleanup and Gathering), two policy-synthesizer LLMs, and two welfare objec-  
8 tives (utilitarian efficiency and Rawlsian maximin), the researcher reliably exceeds  
9 hand-designed baselines, sharply tightens run-to-run variance, and outperforms  
10 prompt-only optimization. The discovered pipelines are objective-dependent: un-  
11 der maximin the researcher independently rediscovers *duty rotation* as a fairness  
12 mechanism, supporting an *information-design* reading in which the researcher  
13 chooses what to reveal to the boundedly rational synthesizer. Code at <https://github.com/anon590/llm-policies-social-dilemmas>.  
14

## 15 1 Introduction

16 Sequential Social Dilemmas (SSDs) [1] are the multi-agent analogue of the prisoner’s dilemma in  
17 temporally rich Markov games: individually rational play leads to collectively suboptimal outcomes  
18 through pollution, over-harvesting, or open conflict. Standard multi-agent reinforcement learning  
19 (MARL) struggles in this regime due to credit assignment, non-stationarity, and large joint action  
20 spaces [2]. A complementary approach, recently introduced by Gallego [3], replaces parameter-  
21 space optimization with *algorithm-space synthesis*: a frozen LLM writes a Python policy function,  
22 evaluates it in self-play, and iteratively refines it from performance feedback. A single generation  
23 step can produce coordination logic (territory partitioning, role assignment, conditional cooperation)  
24 at a sample efficiency several orders of magnitude beyond what gradient-based MARL achieves on  
25 the same environments.

26 This shifts where the design problem lives, rather than removing it. The inner-loop pipeline that  
27 drives the synthesizer has many free parameters: which system prompt, which feedback variables,  
28 which helper functions, how many refinement steps. Each materially affects the resulting policies,  
29 and prior work tuned them by hand. A natural question follows: *can an AI agent design the pipeline?*

30 We answer affirmatively with a two-level autoresearch framework. An *outer-loop researcher agent*  
31 (Claude Opus 4.6, run as a coding agent) edits the source files of an *inner-loop policy synthesizer*  
32 (another LLM), runs evaluations on held-out seeds, and keeps modifications that improve a fixed  
33 welfare objective  $\Phi$ . The outer agent operates on an ordinary git repository (reading code, writing  
34 diffs, running shell commands, etc) without task-specific scaffolding, mirroring the autoresearch  
35 paradigm of Karpathy [4] for single-GPU LLM pretraining. Although the inner-loop SSDs are  
36 gridworld benchmarks rather than physical systems, the outer-loop discovery process itself runs

under conditions a deployed discovery agent faces: noisy multi-seed evaluations, stochastic code generation, an LLM-evaluation budget that bounds how often  $\Phi$  can be queried, and a heterogeneous code repository the agent must navigate end-to-end on its own.

Our contributions are: i) a general two-level framework that delegates the design of an LLM synthesis pipeline to a coding agent operating on a real software repository (Section 3); ii) the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain, with experiments across two SSDs (Cleanup and Gathering), two policy LLMs, and two welfare objectives (utilitarian efficiency  $U$  and Rawlsian maximin  $\min_i R_i$ ) (Section 4); and iii) a mechanism-design interpretation supported by the qualitatively different pipelines the agent produces under different welfare objectives, including the autonomous discovery of *time-based duty rotation* as a fairness mechanism, never found under efficiency optimization.

## 2 Background

We build on the iterative LLM policy synthesis framework of Gallego [3], which serves as the (frozen) inner loop of our two-level system. This section recalls the SSD formalism, the social outcome metrics, and the base synthesis loop.

### 2.1 Sequential Social Dilemmas

A Sequential Social Dilemma is a partially observable Markov game  $\mathcal{G} = \langle N, \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, T, \{R_i\}_{i=1}^N, H \rangle$  with  $N$  agents, state space  $\mathcal{S}$  (the gridworld configuration), per-agent action spaces  $\mathcal{A}_i$ , transition function  $T$ , reward functions  $R_i$ , and episode horizon  $H$  [1]. Beyond the dilemma’s matrix-game structure, SSDs add temporal richness: agents must learn *when* and *where* to cooperate, not just *whether* to.

We study two canonical SSDs that capture complementary dilemma types.

**Cleanup** [5] is a *public goods provision* game ( $N=10$ ). The map has two regions: a river that accumulates waste, and an orchard where apples grow. Apples regrow only when river pollution is below threshold. Each agent can fire a cleaning beam (cost  $-1$ ) that removes waste, collect apples ( $+1$  each), or fire a tagging beam (cost  $-1$ , inflicting  $-50$  on the target and removing it for 25 steps). The dilemma: cleaning is costly but its benefits are public, so purely self-interested agents free-ride.

**Gathering** [1, 6] is a *common pool resource* game ( $N=4$ ). Agents collect shared apples on a fixed respawn timer and may fire tagging beams to temporarily remove rivals. The dilemma: agents can coexist and share resources, or aggress to monopolize them; aggression wastes time and reduces total welfare.

Both games use 8–9 discrete actions (movement, rotation, beam, stand, optionally clean) and episodes of  $H=1000$  steps. The two dilemmas differ in cost structure: asymmetric provision (cleaners pay, all benefit) vs. symmetric restraint (every agent faces the same temptation), a distinction that drives our experimental findings.

**Social metrics.** Following Perolat et al. [6], let  $R_i = \sum_{t=0}^{H-1} r_i^t$  denote agent  $i$ ’s episode return. We evaluate four social outcomes:

$$U = \frac{1}{H} \sum_{i=1}^N R_i \quad (\text{efficiency}) \quad (1)$$

$$E = 1 - \frac{\sum_{i,j} |R_i - R_j|}{2N \sum_i R_i} \quad (\text{equality}) \quad (2)$$

$$S = \frac{1}{N} \sum_{i=1}^N \bar{t}_i \quad (\text{sustainability}) \quad (3)$$

$$P = \frac{1}{H} \sum_{t=0}^{H-1} |\{i : \text{active}_i^t\}| \quad (\text{peace}) \quad (4)$$

where  $\bar{t}_i$  is the mean timestep at which agent  $i$  collects positive reward (higher  $\Rightarrow$  resources preserved later) and  $\text{active}_i^t$  indicates that agent  $i$  is not tagged out at step  $t$ . We additionally consider the

76 *maximin* (Rawlsian) welfare criterion  $\min_i R_i$ , which optimizes the worst-off agent’s return and  
 77 serves as the second objective in our experiments.

## 78 2.2 Iterative LLM Policy Synthesis

79 Let  $\Pi$  denote the space of *code-based policies*: deterministic functions  $\pi : \mathcal{S} \times [N] \rightarrow \mathcal{A}$  expressed  
 80 as executable Python code. Each policy has access to the full environment state and a library of  
 81 helpers (BFS pathfinding, beam targeting, coordinate transforms). This state access is a deliberate  
 82 design choice: programmatic policies operate in algorithm space rather than the reactive observation-  
 83 to-action space of neural policies, which lets a single LLM generation step encode rich coordination  
 84 logic.

85 A frozen LLM  $\mathcal{M}$  acts as the *policy synthesizer*. Given a system prompt  $p$  describing the environ-  
 86 ment API and a feedback prompt  $q_k$ , it produces a new policy

$$\pi_{k+1} = \mathcal{M}(p, q(\pi_k, \mathcal{F}_k)),$$

87 where  $\pi_k$  is the previous policy (its source code) and  $\mathcal{F}_k$  is the evaluation feedback. All  $N$  agents  
 88 execute the same program  $\pi_k$  in self-play. We stress that this is *symmetric*, not *behaviorally ho-*  
 89 *mogeneous*: since  $\pi_k$  takes `agent_id` as an argument, a single shared program can induce distinct  
 90 per-agent behaviors (cleaner vs. gatherer assignment, time-rotated duty cycles, partitioned territo-  
 91 ries; see the synthesized policies in Appendix E). What is shared is the source code; the sampled  
 92 action distributions can differ across agents. Evaluation over a set of random seeds  $S$  yields the  
 93 mean per-agent return  $\bar{r}_k$  and the social metrics vector  $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$ . Each generated  
 94 policy passes an AST-based safety check (blocking eval, file I/O, network access) followed by a  
 95 short smoke test; failures trigger regeneration (up to  $R$  attempts) with the error message appended  
 96 to the prompt.

97 **Feedback.** The hand-designed feedback used by Gallego [3] as our starting point packages the  
 98 previous policy’s source code together with all available evaluation signals:

$$\mathcal{F}_k = (\text{code}(\pi_k), \bar{r}_k, \mathbf{m}_k, \mathbf{d}), \quad (5)$$

99 where  $\mathbf{d}$  contains natural-language definitions of each social metric. Exposing the full social-metrics  
 100 vector (rather than the scalar reward alone) disambiguates distinct failure modes that share the same  
 101 average return (e.g., under- vs. over-cleaning in Cleanup) and consistently improves cooperative  
 102 outcomes. On Cleanup, this baseline feedback reaches  $U=1.93/2.70$  (mean/max) with Gemini and  
 103  $U=0.86/1.56$  with Sonnet across 4 runs each. The choice of feedback content is a single design  
 104 decision within a much larger pipeline configuration space; our two-level framework (Section 3)  
 105 opens the full space to automated search.

## 106 3 Two-Level Framework

107 We introduce a two-level system where a *researcher agent*  $\mathcal{R}$  autonomously discovers configurations  
 108 that optimize the output of an inner-loop system. While we instantiate this for multi-agent policy  
 109 synthesis, the architecture is general: any pipeline where an LLM generates artifacts, evaluates them,  
 110 and iterates can serve as the inner loop. The key insight is that the entire inner-loop codebase is a  
 111 designable artifact that an AI agent can search over. Figure 1 illustrates the architecture.

### 112 3.1 Configuration Space

113 Let  $\mathcal{C}$  denote the space of *pipeline configurations*. Each configuration  $c \in \mathcal{C}$  specifies the full inner-  
 114 loop setup:

$$c = (p, \phi, \mathcal{H}, \iota, v) \quad (6)$$

115 where  $p$  is the system prompt,  $\phi$  is the feedback construction function (which metrics and diagnostics  
 116 to include, how to frame it, whether to inject adaptive hints),  $\mathcal{H}$  is the helper function library,  $\iota$   
 117 specifies the iteration logic (number of iterations  $K$ , sampling strategy, temperature schedule), and  
 118  $v$  is the validation pipeline. Table 1 provides concrete examples.

119 The hand-designed feedback of [3] corresponds to a single fixed instantiation of  $\phi$ . Our framework  
 120 opens the full configuration space to automated search.

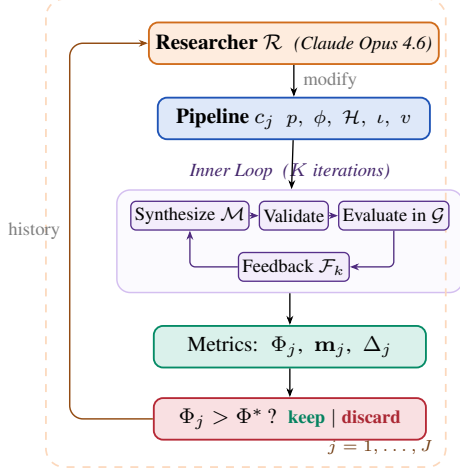


Figure 1: Two-level automated research framework (Algorithm 1).

### Algorithm 1 Two-Level Automated Research

**Require:** Game  $\mathcal{G}$ , policy synthesizer  $\mathcal{M}$ , researcher  $\mathcal{R}$ , system prompt  $p_{\mathcal{R}}$ , initial config  $c_0$ , outer iterations  $J$ , ground-truth objective  $\Phi$ , held-out seeds  $S_{\text{ho}}$

**Ensure:** Best configuration  $c^*$ , best policy  $\pi^*$

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1:  $\pi_0^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_0)$ 
2:  $\Phi_0 \leftarrow \text{EVAL}(\pi_0^*; \mathcal{G}, S_{\text{ho}})$ 
3:  $\text{history} \leftarrow \{(c_0, \Phi_0, \mathbf{m}_0, \emptyset)\}$ 
4: for  $j = 1, \dots, J$  do
5:    $c_j \leftarrow \mathcal{R}(p_{\mathcal{R}}, \text{CODE}(c_{j-1}), \text{history})$ 
6:   if  $\neg \text{VALIDATECONFIG}(c_j)$  then  $\text{retry} \leq R$ 
7:      $\pi_j^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_j)$ 
8:      $\Phi_j \leftarrow \text{EVAL}(\pi_j^*; \mathcal{G}, S_{\text{ho}})$ 
9:      $\Delta_j \leftarrow \text{DIFF}(c_{j-1}, c_j)$  // code diff
10:     $\text{history} \leftarrow \text{history} \cup \{(c_j, \Phi_j, \mathbf{m}_j, \Delta_j)\}$ 
11:  end for
12:  $j^* \leftarrow \arg \max_j \Phi_j$ 
13: return  $c_{j^*}, \pi_{j^*}^*$ 

```

Table 1: Configuration components modifiable by the researcher, with concrete edits that  $\mathcal{R}$  made in our runs. The environment simulator, ground-truth evaluation, and policy LLM weights are frozen.

|               | Scope           | Concrete edits by $\mathcal{R}$ (see appendix)                                                                                                  |
|---------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| $p$           | System prompt   | Multi-section strategic briefing with explicit duty-rotation template ( <code>agent_id + step//T</code> ) % n (App. D.2, Listing 3)             |
| $\phi$        | Feedback fn     | Thresholded diagnostics: “FAIRNESS ALERT” fires when $\min_i R_i < 0$ ; “DO NOT REGRESS” guard fires when $U \geq 2.5$ (App. D.3, Listing 4)    |
| $\mathcal{H}$ | Helper library  | BFS-Voronoi territories with respawn-timer awareness; band-based apple zoning by <code>agent_id</code> (App. D.1, Listings 2, 1)                |
| $\iota$       | Iteration logic | Per-condition $K \in \{2, 3\}$ ; $ S $ walked $5 \rightarrow 8 \rightarrow 12$ for variance control; thinking budget 10–32k (App. D.4, Table 6) |
| $v$           | Validation      | AST blocks <code>eval/file I/O/network</code> ; 50-step smoke test; $\text{retry} \leq R$ on failure                                            |

## 3.2 Inner Loop (Policy Synthesis)

Given a configuration  $c$ , the inner loop executes  $K$  iterations of LLM policy synthesis:

$$\pi_c^* = \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c) \quad (7)$$

Each iteration  $k$  proceeds in four stages, following [3]:

- Synthesize.** The policy LLM  $\mathcal{M}$  receives the system prompt  $p$ , the previous policy’s source code  $\pi_{k-1}$ , and feedback  $\mathcal{F}_{k-1}$  constructed by  $\phi$ . It generates a new Python policy function  $\pi_k$  that has access to full environment state and the helper library  $\mathcal{H}$ .
- Validate.** The generated code undergoes AST-based safety checks (blocking dangerous operations such as file I/O and network access) followed by a short smoke test. Failures trigger re-generation (up to  $R$  retries), with the error message appended to the prompt.
- Evaluate.** All  $N$  agents execute the same policy  $\pi_k$  in self-play over  $|S|$  random seeds (note the policy is conditional on `agent_id`). The evaluation yields the mean per-agent reward  $\bar{r}_k$  and the social metrics vector  $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$ .
- Feedback.** The feedback function  $\phi$  constructs the prompt for the next iteration from  $(\pi_k, \bar{r}_k, \mathbf{m}_k)$ , packaging the previous policy’s source code together with the scalar reward, the social metrics vector, their natural-language definitions, and any adaptive diagnostics that  $\phi$  injects.

The inner loop output is evaluated on held-out seeds via a fixed ground-truth objective:

$$\Phi(c) = \text{EVAL}(\pi_c^*; \mathcal{G}, S_{\text{held-out}}) \quad (8)$$

138 where  $\Phi$  maps from social outcomes to a scalar. We consider two objectives:

$$\Phi_U = U \quad (\text{utilitarian efficiency}) \quad (9)$$

$$\Phi_{\min} = \min_i R_i \quad (\text{Rawlsian maximin}) \quad (10)$$

### 139 3.3 Outer Loop (Automated Research)

140 The researcher agent  $\mathcal{R}$  iteratively modifies the pipeline configuration. Following the autoresearch  
141 paradigm [4],  $\mathcal{R}$  operates on the inner-loop codebase as a modifiable artifact, proposing changes,  
142 observing outcomes, and refining. The procedure is formalized in Algorithm 1.

143 At each outer iteration  $j$ , the researcher  $\mathcal{R}$  receives: i) the **full source code** of the current config-  
144 uration  $c_{j-1}$  (prompts, feedback construction, helpers, iteration logic); ii) the **experiment history**:  
145 for each prior iteration, the code diff  $\Delta_i$ , ground-truth score  $\Phi_i$ , and social metrics vector  $\mathbf{m}_i$ ; iii)  
146 the **environment source code** (read-only), enabling the researcher to reason about game mechanics.  
147 The researcher proposes a new configuration  $c_j$  by generating code modifications. Concretely,  $\mathcal{R}$   
148 is a coding agent (Claude Code CLI) that operates on a real software repository: it reads and edits  
149 Python source files, runs shell commands, inspects evaluation outputs, and commits changes to a  
150 dedicated git branch, following the same workflow a human researcher would follow. This mirrors  
151 the original autoresearch setup [4], where an AI coding agent modifies `train.py` and observes  
152 validation loss; here, the researcher modifies the inner-loop codebase and observes social welfare.

### 153 3.4 Connection to Automated Mechanism Design

154 The two-level structure admits a mechanism design interpretation. The researcher  $\mathcal{R}$  acts as a *mech-*  
155 *anism designer*: it controls the information structure (what metrics to reveal, how to frame them),  
156 the action space (what helper functions are available), and the incentive structure (how feedback is  
157 presented) under which the policy synthesizer  $\mathcal{M}$  operates. The synthesizer acts as the *agent* within  
158 the designed mechanism.

159 This connects to the automated mechanism design literature [7], where a principal designs rules to  
160 induce desired behavior from self-interested agents. In our setting:

- 161 • The **principal** is the researcher  $\mathcal{R}$ , optimizing the ground-truth objective  $\Phi$ .
- 162 • The **agent** is the synthesizer  $\mathcal{M}$ , optimizing per-agent reward as instructed.
- 163 • The **mechanism** is the configuration  $c$ : prompts, feedback, helpers, iteration logic.
- 164 • The **outcome** is the social welfare of the resulting multi-agent policy  $\pi_c^*$ .

165 A crucial distinction from classical mechanism design:  $\mathcal{M}$  follows instructions but has *bounded*  
166 *rationality* in the sense that its ability to synthesize effective policies depends on the information  
167 and tools provided. The researcher’s task is thus closer to *information design* [8]: choosing what  
168 to reveal to help  $\mathcal{M}$  navigate the cooperation–defection tension. Our experiments (Section 4) show  
169 that the researcher designs qualitatively different information structures depending on the welfare  
170 objective  $\Phi$ , supporting this interpretation empirically.

## 171 4 Experiments

### 172 4.1 Setup

173 We conduct 12 autonomous researcher runs across a factorial design. The researcher agent  $\mathcal{R}$  is  
174 Claude Opus 4.6, invoked via the Claude Code CLI as a coding agent. Each run operates on a dedi-  
175 cated git branch of a real Python codebase:  $\mathcal{R}$  edits source files in `pipeline/`, executes evaluation  
176 scripts, reads metric outputs, and iterates, without human intervention.

177 **Design.** For Cleanup: 2 policy LLMs  $\times$  2 objectives  $\times$  2 replications = 8 runs. For Gathering: 2  
178 policy LLMs  $\times$  1 objective  $\times$  2 replications = 4 runs. Maximin runs are unnecessary for Gather-  
179 ing because efficiency optimization alone achieves equality  $E > 0.94$  (the game’s symmetric cost  
180 structure makes fairness a free byproduct of coordination).

181 **Models.** Policy synthesizer  $\mathcal{M}$ : Gemini 3.1 Pro (Google) or Claude Sonnet 4.6 (Anthropic). Both  
182 use extended thinking. These are the same models evaluated in [3]. We additionally test with

Table 2: Results. Mean / max across 2 runs per condition (4 for Cleanup baselines). Baseline: unmodified pipeline from [3]. GEPA: automated prompt optimization [9] with matched compute budget.

| Policy LLM                            | Target        | $U$               | $E$               | $\min_i R_i$    |
|---------------------------------------|---------------|-------------------|-------------------|-----------------|
| <i>Cleanup (N=10) — Baseline</i>      |               |                   |                   |                 |
| Gemini 3.1 Pro                        | —             | 1.93/2.70         | 0.17/0.62         | −159/−84        |
| Sonnet 4.6                            | —             | 0.86/1.56         | −0.02/0.25        | −151/−59        |
| <i>Cleanup (N=10) — GEPA [9]</i>      |               |                   |                   |                 |
| Gemini 3.1 Pro                        | $\Phi_U$      | 1.34/1.37         | −0.10/−0.05       | −149/−126       |
|                                       | $\Phi_{\min}$ | 1.76/2.76         | 0.90/0.96         | 143/245         |
| Sonnet 4.6                            | $\Phi_U$      | 1.04/1.20         | −0.59/−0.21       | −164/−126       |
|                                       | $\Phi_{\min}$ | −0.63/1.60        | 0.77/1.00         | −147/6          |
| <i>Cleanup (N=10) — Autoresearch</i>  |               |                   |                   |                 |
| Gemini 3.1 Pro                        | $\Phi_U$      | <b>3.20</b> /3.25 | 0.55/0.61         | −211/−182       |
|                                       | $\Phi_{\min}$ | 3.16/3.19         | <b>0.98</b> /0.98 | <b>290</b> /296 |
| Sonnet 4.6                            | $\Phi_U$      | <b>3.12</b> /3.14 | 0.66/0.70         | −196/−146       |
|                                       | $\Phi_{\min}$ | 2.57/2.93         | <b>0.91</b> /0.97 | <b>179</b> /200 |
| <i>Gathering (N=4) — Baseline</i>     |               |                   |                   |                 |
| Gemini 3.1 Pro                        | —             | 2.04/2.42         | 0.90/0.98         | 412/571         |
| Sonnet 4.6                            | —             | 0.03/0.03         | 0.54/0.54         | 0/0             |
| <i>Gathering (N=4) — GEPA [9]</i>     |               |                   |                   |                 |
| Gemini 3.1 Pro                        | $\Phi_U$      | 2.08/2.35         | 0.94/0.96         | 436/518         |
| Sonnet 4.6                            | $\Phi_U$      | 1.20/1.23         | 0.63/0.71         | 86/123          |
| <i>Gathering (N=4) — Autoresearch</i> |               |                   |                   |                 |
| Gemini 3.1 Pro                        | $\Phi_U$      | 2.49/2.51         | 0.98/0.98         | 582/582         |
| Sonnet 4.6                            | $\Phi_U$      | 2.52/2.52         | 0.96/0.98         | 576/597         |

183 Gemma 4 26B-A4B-IT (Google), a smaller open-weight model, to probe the framework’s behavior  
 184 when the policy synthesizer has substantially lower capability (Appendix C).

185 **Baselines.** The hand-designed feedback configuration from [3] serves as the initial pipeline  $c_0$   
 186 for all runs. On Cleanup ( $N=10$ ), this baseline achieves  $U=1.93/2.70$  (mean/max) with Gemini  
 187 and  $U=0.86/1.56$  with Sonnet across 4 runs each. We additionally compare against GEPA [9], an  
 188 automated prompt optimization method that iteratively refines the system prompt via LLM reflection.  
 189 GEPA optimizes only the system prompt  $p$ , whereas our researcher modifies the full pipeline  
 190  $(p, \phi, \mathcal{H}, \iota, v)$ . We give GEPA a matched compute budget: the same number of optimization steps  
 191 as outer iterations used by our method (each GEPA step costs  $\sim 3$  policy evaluations, comparable to  
 192 our inner loop’s  $K+1=4$ ).

193 **Inner loop parameters.**  $K=3$  synthesis iterations (or  $K=2$  if the researcher elects), evaluated  
 194 over  $|S|=5-12$  random seeds (researcher-configurable). The researcher ran  $J=3-17$  outer iterations  
 195 per run (mean: 8.3), with a keep rate of 18%–100% (mean: 63%).

## 196 4.2 Results

197 Table 2 presents the main results, comparing autoresearch against the hand-designed baseline of [3]  
 198 and GEPA prompt optimization [9].

199 **Finding 1: The researcher reliably improves over hand-designed baselines and outperforms**  
 200 **prompt-only optimization.** All 12 runs show substantial improvement regardless of starting point  
 201 (Figure 2). In Cleanup, the researcher exceeds the unmodified pipeline baseline by 66% with Gemini  
 202 ( $U: 1.93 \rightarrow 3.20$ ) and 263% with Sonnet ( $U: 0.86 \rightarrow 3.12$ ). Strikingly, the Sonnet improvement  
 203 nearly closes the gap with Gemini ( $U=3.12$  vs. 3.20), despite a  $2\times$  baseline disadvantage (0.86 vs.  
 204 1.93). Final efficiency values cluster tightly around  $U \approx 3.1-3.2$  despite baselines varying from  
 205 0.29 to 2.70 across individual runs, suggesting the researcher reliably discovers the performance  
 206 ceiling of each policy LLM. Beyond improving the mean, the researcher yields remarkably consistent  
 207 results: under efficiency optimization, Gemini’s runs cluster within  $U=3.20-3.25$  (gap 0.05)

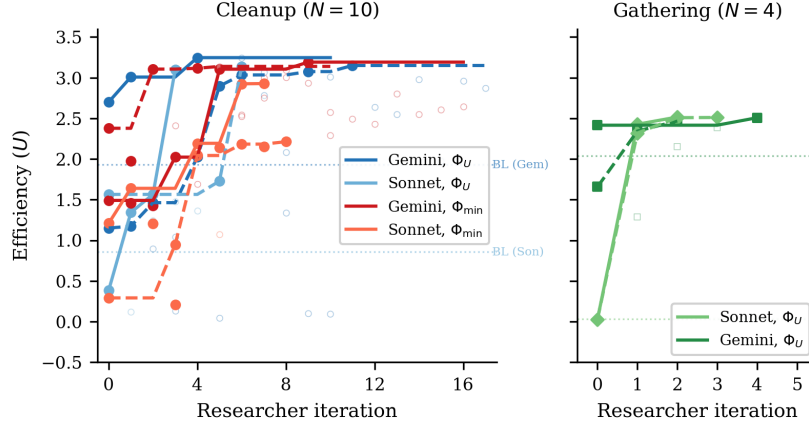


Figure 2: Efficiency ( $U$ ) across researcher iterations for all 12 runs. *Left*: Cleanup ( $N=10$ ) with 8 runs across 2 LLMs  $\times$  2 objectives (solid lines = kept experiments, open circles = discarded). Dashed horizontal line: hand-designed baseline from [3]. All runs converge to  $U \approx 3.1$ – $3.2$  despite diverse starting points. *Right*: Gathering ( $N=4$ ) with 4 runs (2 per LLM). All converge to  $U \approx 2.5$  within 2–4 iterations despite baselines ranging from 0.03 to 2.42.

despite baselines spanning 1.15–2.70, and Sonnet’s within 3.12–3.14 (gap 0.02) from baselines of 0.38–1.56. In Gathering, all four runs converge to  $U=2.47$ – $2.52$  from baselines ranging 0.03–2.42. The researcher discovers pipeline modifications: structured helpers (App. D.1), strategic hints (App. D.2), tailored feedback (App. D.3). These constrain the policy LLM toward consistently good solutions.

Compared to GEPA (Table 2), autoresearch achieves  $2.4\times$  higher efficiency under  $\Phi_U$  in Cleanup ( $U=3.20$  vs. 1.34),  $2\times$  higher maximin under  $\Phi_{\min}$  (290 vs. 143), and 20% higher efficiency in Gathering ( $U=2.49$  vs. 2.08). Critically, GEPA shows far wider spread across runs on the harder fairness objective ( $\Phi_{\min}$  maximin in Cleanup: 245/41 vs. autoresearch’s 296/284). The gap widens further with Sonnet: on Cleanup  $\Phi_U$ , GEPA stalls at  $U=1.04$  ( $3\times$  below autoresearch’s 3.12), and on  $\Phi_{\min}$  its two runs split between a modestly positive solution ( $\min_i R_i=6$ ,  $U=1.60$ ) and a pathological “everyone cleans, nobody eats” regime (perfect equality  $E=1.00$  but negative efficiency  $U=-2.87$  and  $\min_i R_i=-299$ ), while autoresearch with the same model reaches  $\min_i R_i=179/200$  reliably. This highlights the advantage of modifying the full pipeline, whereas prompt-only optimization cannot address the underlying stochasticity of code generation.

**Finding 2: No efficiency–fairness tradeoff in Cleanup (Gemini).** Maximin-optimized Gemini pipelines sacrifice only 1% efficiency ( $U$ : 3.16 vs. 3.20) while achieving near-perfect equality ( $E$ : 0.98 vs. 0.55) and transforming maximin from deeply negative baselines (−99 to −84) to  $\min_i R_i = 290$ . The researcher discovers *fair duty rotation* (Listing 6; primed by the prompt rewrite of Listing 3)—time-based cycling using `agent_id` and `env._step_count`—simultaneously improving worst-off welfare *and* collective output. Because cleaning is a public good, distributing the cleaning cost fairly ensures enough cleaners to sustain apple production.

For Sonnet, there is a moderate tradeoff: efficiency drops from 3.12 to 2.57 under maximin optimization. The gap reflects Sonnet’s harder time implementing complex coordination mechanisms (role rotation, zone assignment) from strategic hints alone.

**Finding 3: Game structure determines whether fairness requires explicit optimization.** In Cleanup, where cleaning costs are borne asymmetrically (cleaners pay −1, free-riders collect apples), baseline equality ranges from  $E=0.04$  to 0.62, and maximin optimization is required to reach  $E > 0.9$ . In Gathering, where all agents face a symmetric landscape, efficiency optimization alone achieves  $E > 0.94$  across all 4 runs: no separate maximin runs are needed.

This generalizes: in *provision* dilemmas with asymmetric costs, fairness requires designed mechanisms (role rotation, duty sharing); in *restraint* dilemmas with symmetric costs, fairness emerges as a

free byproduct of efficient coordination. The researcher independently discovers this, it creates role differentiation pipelines only for Cleanup, and pure spatial-coordination pipelines for Gathering.

**Finding 4: Convergent discovery of qualitatively different strategies per objective.** Despite fully independent runs, the researcher converges on the same core strategies within each condition (Table 3, Appendix A). In Cleanup, waste-counting helpers and spatial zone partitioning appear across all 8 runs. The key qualitative difference is *role rotation*, appearing in 4/4 maximin runs (Listings 6, 7) but 0/4 efficiency runs (Listing 5). Under efficiency optimization, the researcher assigns static cleaning roles (some agents always clean), achieving high collective output at the cost of equality. Under maximin optimization, it discovers rotation as a fairness mechanism.

In Gathering, the researcher discovers BFS-Voronoi territory partitioning and respawn-timer awareness (Listings 2, 8), with no role differentiation. Thus, optimization is purely spatial (who collects which apples) and temporal (respawn-aware positioning).

**Common failure modes.** Several patterns led to discarded experiments across all runs: (1) *over-prescription*: adding too many strategic hints confuses the policy LLM, causing generation failures; (2) *iteration regression*:  $K=4$  or  $K=5$  inner iterations often cause catastrophic regression as the LLM “over-refines” a working policy; (3) *feedback overload*: verbose per-agent diagnostics cause over-correction rather than gradual improvement; (4) *variance exposure*: identical configurations can yield  $U=3.2$  then  $U=0.09$  on different seeds, requiring multi-seed evaluation for stable signals (per-run  $K, |S|$  in Table 6).

## 5 Related Work

**Sequential social dilemmas.** SSDs were introduced by Leibo et al. [1] (Gathering) and extended to public-goods settings by Hughes et al. [5] (Cleanup); Perolat et al. [6] formalized the social outcome metrics ( $U, E, S, P$ ) used here. Subsequent work studies inequity aversion, intrinsic motivation, and reputational mechanisms for promoting cooperation in MARL agents. We complement this line by automating the search for cooperative *programs* rather than evolving neural policies, and by treating the welfare objective  $\Phi$  as a designable parameter that the outer agent optimizes for, obtaining qualitatively different cooperative behaviors (static role assignment vs. duty rotation) under different objectives without changing the environment.

**LLMs for policy and program synthesis.** FunSearch [10] evolves programs for combinatorial discovery; Eureka [11] synthesizes reward functions from environment source code; Voyager [12] and Code as Policies [13] generate executable skill code for single-agent embodied control; ReEvo [14] evolves heuristics with reflective feedback. These works target single-agent settings or non-strategic optimization. Gallego [3], on which our inner loop is based, extended LLM program synthesis to the multi-agent SSD setting, where one program must coordinate  $N$  self-play copies. Our contribution is one level above: rather than tuning the synthesizer for one task, we let an outer agent rewrite the synthesis pipeline itself.

**LLM reflection and prompt optimization.** Reflexion [15], Self-Refine [16], OPRO [17], and GEPA [9] demonstrate that structured verbal feedback loops improve LLM outputs; ERL [18] internalizes such reflection via self-distillation. These methods optimize the prompt or the model’s reasoning trajectory in isolation. Our framework instead modifies the entire surrounding pipeline (system prompt, feedback construction, helper library, iteration logic, validation) making prompt optimization one component of a strictly larger search space. The empirical gap is large: Section 4 shows full-pipeline autoresearch achieving  $3\times$  higher efficiency in Cleanup and 37% higher in Gathering than GEPA at matched compute, with run-to-run spread an order of magnitude tighter on the harder fairness objective.

**Automated AI research.** A small but growing body of work delegates the design of ML pipelines to LLM coding agents. Karpathy’s autoresearch [4] runs a coding agent that modifies `train.py` for nanoGPT pretraining and is rewarded for validation loss. Our system shares this autoresearch architecture (coding agent + frozen evaluation harness + diff-based history) but targets a different inner loop (multi-agent policy synthesis instead of single-model pretraining) and a different objective

(multi-agent social welfare instead of validation bits-per-byte). To our knowledge this is the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain.

**Automated mechanism and information design.** Classical automated mechanism design [7] computes optimal allocation rules for self-interested agents under explicit incentive constraints, while information design [8] studies how a principal commits to a signaling policy that shapes a receiver’s behavior. Our outer agent solves a related but distinct problem:  $\mathcal{M}$  is not strategically deceptive (it follows instructions) but is *boundedly rational*, so the researcher must decide what information, helpers, and structure to expose so that  $\mathcal{M}$  writes policies achieving the principal’s welfare goal. Section 4 shows that the pipelines  $\mathcal{R}$  produces are genuinely a function of the welfare objective, supporting this information-design framing empirically.

## 6 Discussion and Conclusion

We have presented a two-level framework where an AI coding agent autonomously discovers pipeline configurations that improve the output of an inner-loop LLM system. Applied to multi-agent policy synthesis in social dilemmas, the researcher agent reliably exceeds hand-designed baselines across 12 independent runs, and converges on qualitatively similar strategies within each game-objective condition. The framework requires no domain-specific scaffolding: the agent operates on a standard software repository using the same tools (file editing, shell commands, git) available to a human researcher.

**Mechanism design in action.** Our results empirically support the mechanism design interpretation of Section 3.4. The researcher acts as an information designer: under  $\Phi_U$ , it reveals efficiency-oriented information (waste counts, zone assignments) that guides  $\mathcal{M}$  toward productive but unequal role allocation (Listing 5). Under  $\Phi_{\min}$ , it additionally reveals fairness-oriented structure such as rotation schedules and per-agent equity feedback (Listings 3, 4), guiding  $\mathcal{M}$  toward egalitarian coordination. The striking absence of a fairness–efficiency tradeoff with Gemini suggests that in public goods provision, the mechanism design problem has a cooperative optimum where both objectives align.

**Two types of dilemma.** The Cleanup–Gathering contrast reveals a structural insight about social dilemmas: *provision* dilemmas (asymmetric costs, one agent’s contribution benefits all) require explicit fairness mechanisms, while *restraint* dilemmas (symmetric costs, each agent faces the same temptation) achieve fairness naturally when coordination improves. This distinction, well-known in the common-pool resource literature, is rediscovered here by an autonomous AI agent, which suggests it is a robust structural feature of these environments.

**Human oversight and audit.** Our system is fully autonomous by design, but its architecture offers natural affordances for human-in-the-loop oversight. Because the researcher  $\mathcal{R}$  operates on a standard git repository, a domain expert can audit the full history of code diffs  $\Delta_j$  alongside their metric outcomes  $(\Phi_j, \mathbf{m}_j)$ , exactly as they would review a colleague’s pull requests. The keep/discard decision (Algorithm 1, line 11) is a lightweight intervention point: a human could override the accept threshold, veto modifications that introduce undesirable mechanisms (e.g., exploitative role assignments), or steer the researcher toward under-explored regions of the configuration space by editing the experiment history. More broadly, the separation between the researcher  $\mathcal{R}$  and the frozen evaluation  $\Phi$  creates a *delegation boundary*: the human defines *what* to optimize (the welfare objective), while the agent decides *how*. This mirrors the expert-in-the-loop design patterns identified in deployment-oriented work on AI agents for discovery, where trust calibration depends on the legibility of the agent’s decision trail rather than on real-time supervision.

**Future work.** Several directions emerge. First, we are interested in applying the two-level framework to other LLM-driven pipelines (code optimization, scientific experiment design, infrastructure tuning), where the same architecture applies with a different inner loop and objective  $\Phi$ ; next, adversarial objectives can be intriguing, to test whether the researcher discovers exploitative pipeline configurations, exposing reward hacking risks; and asymmetric programs: extend to settings where agents run *different* source code (the present setup is symmetric in code but already supports hetero-

geneous behavior via `agent_id`; relaxing the shared-program constraint enables richer per-agent specialization and partial-information settings).

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## A Additional Results

Table 3: Strategies discovered by the researcher in Cleanup across 4 efficiency-optimized and 4 maximin-optimized independent runs.

| Strategy                        | $\Phi_U$ (4 runs) | $\Phi_{\min}$ (4 runs) |
|---------------------------------|-------------------|------------------------|
| Waste-counting helpers          | 4/4               | 4/4                    |
| Zone/lane partitioning          | 3/4               | 4/4                    |
| Anti-regression feedback        | 3/4               | 2/4                    |
| Worked policy examples          | 2/4               | 3/4                    |
| Cleaning cost economics         | 2/4               | 3/4                    |
| <b>Time-based role rotation</b> | <b>0/4</b>        | <b>4/4</b>             |

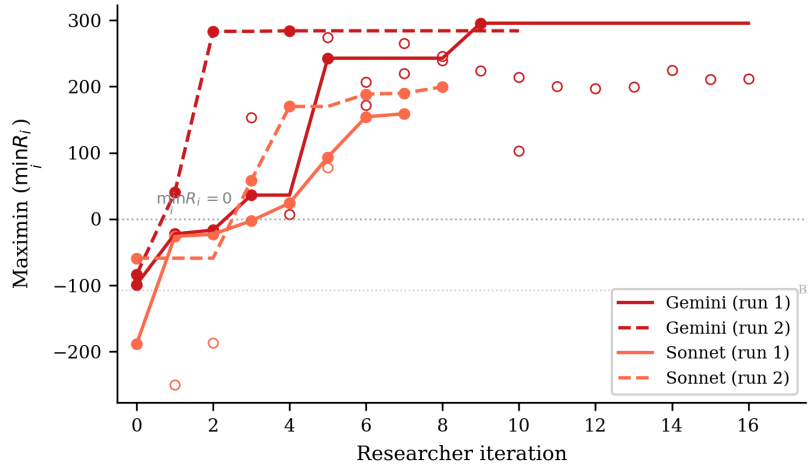


Figure 3: Maximin ( $\min_i R_i$ ) across researcher iterations for the 4 Cleanup maximin-optimized runs. All runs transform deeply negative baselines (worst-off agents losing reward) into substantially positive values. Gemini reaches  $\sim 290$  while Sonnet reaches  $\sim 160$ – $200$ . The dashed line marks  $\min_i R_i = 0$  (no agent loses reward overall).

## B Compute Requirements

Table 4 reports wall-clock time for all 12 autonomous runs and the 6 GEPA comparison runs. *Inner loop* measures total time spent on policy LLM generation and environment simulation across all evaluations; *total* (estimated from run-directory timestamps) additionally includes the researcher agent’s analysis, code editing, and decision-making between evaluations.

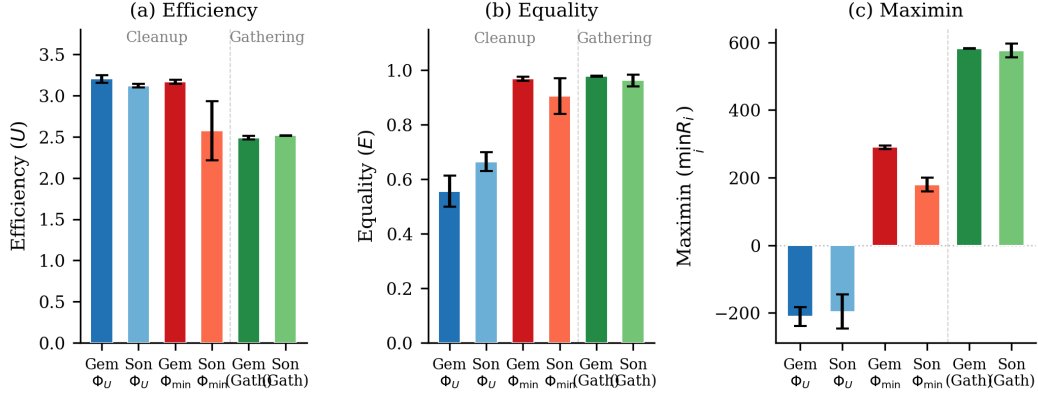


Figure 4: Final metrics across all conditions. (a) Efficiency: all conditions converge to  $U \approx 2.5$ – $3.2$ . (b) Equality: maximin-optimized Cleanup runs achieve  $E \approx 1.0$ , while efficiency-optimized runs show  $E \approx 0.5$ – $0.7$ ; Gathering achieves high equality regardless. (c) Maximin: the sharpest contrast—efficiency optimization leaves the worst-off agent deeply negative ( $\min_i R_i \approx -200$ ), while maximin optimization transforms it to  $+290$  (Gemini) or  $+179$  (Sonnet). Gathering achieves high maximin ( $\sim 580$ ) under efficiency optimization alone. Error bars show s.d. across runs.

Table 4: Wall-clock time per autonomous run. Evals: total inner loop evaluations including initial baseline. Per eval: mean wall-clock per single evaluation. Total: end-to-end including researcher overhead.

| Policy LLM                 | $\Phi$        | Evals | Inner loop (h) | Per eval (min) | Total (h) |
|----------------------------|---------------|-------|----------------|----------------|-----------|
| <i>Cleanup</i> ( $N=10$ )  |               |       |                |                |           |
| Gemini                     | $\Phi_U$      | 11    | 2.6            | 14             | 3.7       |
| Gemini                     | $\Phi_U$      | 18    | 4.1            | 14             | 6.4       |
| Sonnet                     | $\Phi_U$      | 4     | 2.1            | 32             | 6.1       |
| Sonnet                     | $\Phi_U$      | 7     | 5.0            | 43             | 6.1       |
| Gemini                     | $\Phi_{\min}$ | 17    | 3.0            | 11             | 4.3       |
| Gemini                     | $\Phi_{\min}$ | 11    | 3.6            | 20             | 5.0       |
| Sonnet                     | $\Phi_{\min}$ | 8     | 4.5            | 33             | 8.8       |
| Sonnet                     | $\Phi_{\min}$ | 9     | 5.3            | 36             | 13.2      |
| <i>Gathering</i> ( $N=4$ ) |               |       |                |                |           |
| Sonnet                     | $\Phi_U$      | 3     | 1.7            | 33             | 2.2       |
| Gemini                     | $\Phi_U$      | 5     | 1.4            | 17             | 1.9       |
| Gemini                     | $\Phi_U$      | 3     | 0.9            | 19             | 1.1       |
| Sonnet                     | $\Phi_U$      | 4     | 2.3            | 35             | 3.4       |
| All 12 runs                |               | 100   | 36.6           | 22             | 62.2      |
| GEPA (6 runs)              |               | 6     | 5.0            | 50             | —         |

**Cost breakdown.** Each inner loop evaluation involves  $K=2$ – $3$  policy LLM generation calls (each  $\sim 2k$  input tokens,  $\sim 1.5k$  output tokens) followed by multi-seed simulation (5–12 seeds,  $\sim 15$ – $30s$  total). Policy LLM generation dominates inner loop time (86–97%), with Sonnet evaluations taking  $\sim 2\times$  longer than Gemini due to extended thinking. The researcher agent (Claude Opus 4.6, running via Claude Code CLI) accounts for 41% of total wall-clock time (25.6h of the 62.2h total across all 12 runs), spent reading results, editing pipeline source files, and planning modifications. In monetary terms, the researcher agent is the dominant cost: each outer iteration consumes  $\sim 50$ – $100k$  context tokens in the Opus session, whereas each inner loop evaluation uses only  $\sim 5$ – $10k$  policy LLM tokens. The 6 GEPA comparison runs required 5.0h of compute with no researcher agent, operating at lower cost per run but achieving substantially lower performance (Table 2).

## C Smaller Open-Weight Model: Gemma 4 26B

We test the framework with Gemma 4 26B-A4B-IT (Google), a 26B-parameter open-weight model substantially smaller than the frontier models in the main experiments. We run three experiments—two on Cleanup ( $N=10$ ) optimizing efficiency and maximin respectively, and one on Gathering ( $N=4$ ) optimizing efficiency—all with Opus 4.6 as the researcher  $\mathcal{R}$ .

**Setup.** In all three experiments, Gemma starts from complete failure: the baseline pipeline produces  $U=-10.0$  on Cleanup (agents CLEAN-spam without collecting apples) and  $U=0.0$  on Gathering (broken BFS calls), compared to  $U=1.93/0.86$  (Gemini/Sonnet on Cleanup) and  $U=2.04/0.03$  (Gemini/Sonnet on Gathering). The researcher ran  $J=10-18$  outer iterations per condition.

**Results.** Table 5 presents the best configurations discovered. Under efficiency optimization, the researcher achieves  $U=0.87$ —a dramatic recovery from the broken baseline but far below frontier models ( $U \approx 3.2$ ). The researcher compensates for the model’s limited capability by reducing inner-loop iterations to  $K=1$  (avoiding the regression that plagued  $K \geq 2$  runs) and providing highly structured worked examples with explicit cleaning-role assignment.

Table 5: Autoresearch with Gemma 4 26B-A4B-IT. Single run per condition. Baseline: pipeline from [3].

| Game      | Target        | $U$         | $E$               | $\min_i R_i$ | Keep rate |
|-----------|---------------|-------------|-------------------|--------------|-----------|
| Cleanup   | Baseline      | -10.0       | 1.00 <sup>†</sup> | -1000        | —         |
|           | $\Phi_U$      | 0.87        | -1.11             | -371         | 6/19      |
|           | $\Phi_{\min}$ | <b>1.71</b> | <b>0.94</b>       | <b>137</b>   | 9/17      |
| Gathering | Baseline      | 0.0         | 1.00 <sup>†</sup> | 0            | —         |
|           | $\Phi_U$      | <b>2.44</b> | <b>0.98</b>       | <b>580</b>   | 4/11      |

<sup>†</sup>Equality is trivially 1.0 because all agents receive near-zero reward.

**Maximin optimization rescues efficiency.** Strikingly, under maximin optimization the researcher achieves *higher* efficiency ( $U=1.71$ ) than under direct efficiency optimization ( $U=0.87$ ), alongside strong equality ( $E=0.94$ ) and positive worst-off welfare ( $\min_i R_i=137$ ). This reversal—absent in frontier models, where both objectives yield  $U \approx 3.1-3.2$ —occurs because the maximin objective forces the researcher toward coordination mechanisms (rotating cleaning duties, index-based apple assignment) that simultaneously improve collective output. Under efficiency-only optimization, the researcher converges on a local optimum—static role assignment with 25% dedicated cleaners—that a 26B model can implement but that caps performance well below the frontier.

This suggests that *for models below a capability threshold, fairness objectives may serve as better optimization targets for overall social welfare than direct efficiency maximization*. The structured coordination enforced by the maximin objective provides a scaffold that compensates for the weaker model’s difficulty implementing complex strategies from strategic hints alone.

**Gathering: full recovery on a simpler game.** On Gathering ( $N=4$ ), the researcher brings Gemma from  $U=0.0$  to  $U=2.44$  with  $E=0.98$  and  $\min_i R_i=580$ , after 10 outer iterations (4 kept). These values nearly match frontier models (Gemini:  $U=2.49$ , Sonnet:  $U=2.52$ ; Table 2). The researcher discovers the same Voronoi partitioning and respawn-aware camping strategies found in the main experiments, and increases inner-loop iterations to  $K=5$  to allow sufficient refinement. The contrast with Cleanup—where Gemma peaks at  $U=0.87/1.71$  versus frontier  $U \approx 3.2$ —suggests that the researcher can fully compensate for model weakness on simpler coordination tasks (4 agents, symmetric costs) but not on harder provision dilemmas (10 agents, asymmetric costs).

## D Researcher-Authored Pipeline Artifacts

The previous appendix shows the policies  $\pi_c^*$  that the inner loop *outputs*; this one shows the configuration  $c = (p, \phi, \mathcal{H}, \iota, v)$  (Section 4, Table 1) that the researcher  $\mathcal{R}$  *authored* to produce them. Excerpts are taken from the final commit on the dedicated git branch of the corresponding run; we

reproduce the prose verbatim, with author commentary in [brackets] and elisions marked .... We organize by artifact type so each subsection makes a within-type contrast (e.g.,  $\Phi_U$  vs.  $\Phi_{\min}$ , or Cleanup vs. Gathering).

## D.1 Helper library $\mathcal{H}$ : coordination primitives

The researcher adds primitives that the policy LLM can call as black boxes, sparing it from re-implementing tricky logic on every iteration. Two patterns dominate: (i) *state inspection* (waste counts, fractions, beam-yield scoring) and (ii) *coordination primitives* (zone assignment, role rotation). We show one of each.

Listing 1: Cleanup,  $\Phi_{\min}$  – band-based apple zoning helper added by the researcher in exp5. This is the primitive that lets the policy in Listing 6 keep collectors within their own row band, preventing all 7 gatherers from racing to the same apple.

```

458 1 def get_my_apples(env, agent_id):
459 2     """Alive apples in this agent's horizontal row band.
460 3     Divides the orchard into n_agents bands; falls back to all apples if empty."""
461 4     aid = int(agent_id)
462 5     n = int(env.n_agents)
463 6     band_h = env.height / n
464 7     band_lo, band_hi = aid * band_h, (aid + 1) * band_h
465 8
466 9     my_apples, all_apples = set(), set()
467 10    for i in range(env.n_apples):
468 11        if env.apple_alive[i]:
469 12            ar = int(env._apple_pos[i, 0]); ac = int(env._apple_pos[i, 1])
470 13            all_apples.add((ar, ac))
471 14            if band_lo <= ar < band_hi:
472 15                my_apples.add((ar, ac))
473 16    return my_apples if my_apples else all_apples
474

```

Listing 2: Gathering – BFS-Voronoi territory + respawn-aware waiting helpers added by the researcher in gather-exp3. These are the primitives that Listing 8 composes into a one-line policy: “go to my\_zone\_apples, otherwise nearest\_respawning\_apple.”

```

476 1 def voronoi_zones(env):
477 2     """Multi-source BFS Voronoi over walkable cells.
478 3     Returns (row, col) -> agent_id; ties broken by lower agent_id."""
479 4     queue, visited = deque(), {}
480 5     for a_id in range(env.n_agents):
481 6         if int(env.agent_timeout[a_id]) == 0:
482 7             ar = int(env.agent_pos[a_id][0]); ac = int(env.agent_pos[a_id][1])
483 8             queue.append((ar, ac, a_id))
484 9             visited[(ar, ac)] = a_id
485 10    while queue:
486 11        r, c, a_id = queue.popleft()
487 12        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
488 13            nr, nc = r + dr, c + dc
489 14            if 0 <= nr < env.height and 0 <= nc < env.width \
490 15                and not env.walls[nr, nc] and (nr, nc) not in visited:
491 16                visited[(nr, nc)] = a_id
492 17                queue.append((nr, nc, a_id))
493 18    return visited
494
495 19
496 20 def nearest_respawning_apple(env, agent_id, zones=None, max_timer=10):
497 21     """Nearest dead apple in MY zone respawning within max_timer steps.
498 22     Returns (row, col) of best wait spot, or None."""
499 23     if zones is None: zones = voronoi_zones(env)
500 24     best_pos, best_t, best_d = None, max_timer + 1, float('inf')
501 25     ar = int(env.agent_pos[agent_id][0]); ac = int(env.agent_pos[agent_id][1])
502 26     for i in range(env.n_apples):
503 27         if not env.apple_alive[i]:
504 28             pos = (int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
505 29             if zones.get(pos) == agent_id:
506 30                 t = int(env.apple_timer[i])
507 31                 if t <= max_timer:
508 32                     d = abs(pos[0] - ar) + abs(pos[1] - ac)
509 33                     if t < best_t or (t == best_t and d < best_d):
510 34                         best_pos, best_t, best_d = pos, t, d
511 35    return best_pos
512

```

The choice of helper depends on the dilemma’s geometry: row-bands suffice when fairness is the main concern (Cleanup maximin) but full BFS-Voronoi is needed when wall-aware territory owner-

515 ship matters (Gathering). The researcher does not fix this in advance — it picks the simpler primitive  
 516 whenever it works.

## 517 D.2 System prompt $p$ : from neutral framing to strategic briefing

518 The unmodified system prompt is a neutral API description: “Write a policy that maximizes per-  
 519 agent reward.” Under maximin optimization, the researcher rewrites the prompt into a multi-section  
 520 strategic briefing. The excerpt below shows the pieces it added to the Cleanup prompt in exp5; the  
 521 full prompt grew from 165 to 325 lines.

Listing 3: Cleanup,  $\Phi_{\min}$  – excerpts from the researcher-rewritten system prompt (exp5). The unmodified baseline contained only API documentation; everything below was added by  $\mathcal{R}$ .

```

522 ## CRITICAL OBJECTIVE: Rawlsian Fairness (Maximin)
523
524 Your goal is to maximize the minimum per-agent total return across all
525 agents. This is the "maximin" or Rawlsian welfare criterion.
526
527 This means: it is NOT enough for the *average* agent to do well. The
528 worst-off agent must do as well as possible. A policy where 8 agents
529 each earn +200 but 2 agents each earn -100 is TERRIBLE (maximin = -100).
530
531 Key implication: cleaning costs (-1 per CLEAN action) must be shared
532 equitably among ALL agents. If you assign fixed "cleaner" roles, those
533 agents will accumulate large negative rewards and destroy your maximin score.
534
535 ## Strategy for Maximin: ROLE ROTATION + APPLE ZONING
536
537 The optimal strategy for maximin involves:
538 1. Shared cleaning duty: ALL agents take turns cleaning using a rotation
539    schedule based on 'agent_id' and 'env._step_count'. For example:
540    'is_my_cleaning_turn = (agent_id + env._step_count // SHIFT) % env.n_agents < NUM_CLEANERS'
541    where SHIFT is ~50 steps and NUM_CLEANERS is 2-3.
542 2. When NOT your turn: collect apples from YOUR ZONE using
543    'get_my_apples(env, agent_id)' -- prevents all gatherers competing
544    for the same nearest apple.
545 3. NEVER use BEAM (action 6): it costs -1 and causes -50 to the target.
546 4. Keep waste density low: aim for ~2-3 active cleaners at any time.
547
548 [... API documentation, helper documentation, working example ...]
549
550 IMPORTANT:
551 - NEVER assign permanent cleaning roles to specific agents -- this kills maximin.
552 - Use env._step_count for time-based role rotation so all agents share cleaning duty.
553 - NEVER use BEAM (action 6) -- it destroys both agents' rewards.
554
555 
```

556 Two things stand out. First, the researcher writes the formula it expects  $\mathcal{M}$  to use almost verbatim —  
 557  $(\text{agent\_id} + \text{env}.\_step\_count // \text{SHIFT}) \% \text{env}.\_n\_agents < \text{NUM\_CLEANERS}$  — and the  
 558 policy in Listing 6 uses essentially this template. Second, the researcher *teaches by counter-example*:  
 559 the explicit “8 agents earn +200, 2 earn -100, maximin = -100” makes the failure mode of  $\Phi_U$ -style  
 560 static roles (Listing 5) concretely visible to  $\mathcal{M}$ . This is the information-design move predicted by  
 561 the mechanism-design framing in Section 3.4.

562 The Gathering prompt evolves along a different axis (no maximin, no rotation). Its researcher-added  
 563 “Key Strategic Insights” section instead emphasizes (i) *self-play implies never beam* and (ii) *Voronoi*  
 564 *partition each step*, exactly matching what the policy in Listing 8 implements.

## 565 D.3 Feedback $\phi$ : adaptive diagnostics

566 The baseline feedback function from [3] shows reward + four social metrics with definitions and  
 567 stops there. The researcher turns it into a state machine: it inspects the latest metrics and condi-  
 568 tionally injects different hints. Two different runs produced two qualitatively different diagnostics  
 569 — the same metric ( $\bar{r}_k$  or  $\min_i R_i$ ) is repurposed to either *stabilize* a working policy or *redirect* a  
 570 failing one.

Listing 4: Adaptive feedback diagnostics. *Top*: efficiency-optimized run (exp1) – a stability guard prevents iter-3 regression once  $U$  is high. *Bottom*: maximin-optimized run (exp5) – two fairness diagnostics that fire when the worst-off agent loses reward.

```

571 1 # --- Cleanup, Phi_U feedback (exp1, pipeline/feedback.py final) ---
572 2 last_eff = history[-1]["metrics"]["efficiency"]
573

```

```

574 3 if last_eff >= 2.5:
575 4     parts.append(
576 5         """**CRITICAL -- DO NOT REGRESS**: The current policy achieves high "
577 6         "efficiency (>=2.5). You MUST output a policy that is nearly identical "
578 7         "to the current one. Copy the current policy code and make AT MOST "
579 8         "one small targeted improvement (e.g., adjust a single numeric "
580 9         "threshold). If you are not confident a change will help, output the "
581 10        "current policy UNCHANGED. A regression here means the run fails.")
582 11
583 12 # --- Cleanup, Phi_min feedback (exp5, pipeline/feedback.py final) ---
584 13 last_maximin = history[-1]["metrics"].get("maximin", 0)
585 14 last_avg      = history[-1]["reward_avg"]
586 15 if last_maximin < 0:
587 16     parts.append(
588 17         f"""**FAIRNESS ALERT**: maximin={last_maximin:.1f} is NEGATIVE. "
589 18         f"The worst-off agent lost reward overall while average was {last_avg:.1f}. "
590 19         "This means cleaning duties are NOT shared equitably. "
591 20         "Use time-based role rotation (env._step_count) so ALL agents share "
592 21         "cleaning costs equally. NEVER assign permanent cleaner roles.")
593 22 elif last_maximin < last_avg * 0.5:
594 23     parts.append(
595 24         f"""**FAIRNESS WARNING**: maximin={last_maximin:.1f} is much lower than "
596 25         f"average={last_avg:.1f}. The gap suggests unequal cleaning burden. "
597 26         "Ensure ALL agents rotate through cleaning duty.")

```

599 The two diagnostics are written by independent researcher runs but converge on a common pattern:  
 600 *thresholded interventions on a primary metric*. They are also a common cause of the low run-to-  
 601 run variance reported in Finding 1: when a working policy lands in the high- $U$  basin, the stability  
 602 guard prevents the LLM from over-refining and tipping out of it (the  $K=3$  regressions visible in  
 603 Section 4’s “Common failure modes” (4)), and when an unfair policy lands in the negative-maximin  
 604 region, the alert injects exactly the rotation hint that recovers it.

#### 605 **D.4 Iteration logic $\iota$ : per-condition hyperparameters**

606 Although  $\iota$  has the smallest source surface, the researcher’s edits to it explain a meaningful fraction  
 607 of the variance reduction noted in Finding 1. Table 6 summarizes the values shipped in the final  
 608 commit of each best-performing run.

Table 6: Iteration-logic ( $\iota$ ) values in the final commit of selected runs.  $K$  = inner-loop iterations,  
 $|S|$  = evaluation seeds, “thinking” = extended-thinking token budget passed to  $\mathcal{M}$ . Baseline ([3]) is  
 $K=3$ ,  $|S|=5$ , thinking 16k.

| Run                  | Game / objective       | $K$      | $ S $     | thinking   | Researcher’s stated rationale                                           |
|----------------------|------------------------|----------|-----------|------------|-------------------------------------------------------------------------|
| exp1 (Gemini)        | Cleanup, $\Phi_U$      | 3        | 5         | 16k        | default; stability comes from feedback hint                             |
| exp4 (Sonnet)        | Cleanup, $\Phi_U$      | 3        | 5         | 32k        | “Sonnet was over-refining at 16k thinking”                              |
| exp5 (Gemini)        | Cleanup, $\Phi_{\min}$ | <b>2</b> | <b>12</b> | 16k        | “ $K=2$ avoids iter-3 regression; 12 seeds for variance control”        |
| exp7 (Sonnet)        | Cleanup, $\Phi_{\min}$ | 3        | 5         | <b>10k</b> | “reduced thinking from 16k – Sonnet was generating runaway code at 16k” |
| gather-exp3 (Gemini) | Gathering, $\Phi_U$    | 3        | 5         | 16k        | default; Gathering converges in $K \leq 3$                              |

609 The maximin Gemini run is the most informative case: the researcher discovered that with  $K=3$   
 610 and  $|S|=5$ , identical configurations could yield  $\min_i R_i=295$  on one set of seeds and  $\min_i R_i=100$   
 611 on another (this is the “variance exposure” failure mode of Section 4’s common failures). It then  
 612 walked  $|S|$  from  $5 \rightarrow 8 \rightarrow 12$  over three outer iterations until the seed-averaged signal was stable  
 613 enough that the policy LLM stopped chasing noise. The policy in Listing 6 is sampled at  $|S|=12$ .

#### 614 **D.5 Cross-artifact observations**

- 615 • *Helpers do the algebra; prompts pick the strategy class*. The researcher uses helpers (1, 2) to put  
 616 non-trivial primitives at  $\mathcal{M}$ ’s fingertips, and the prompt (3) to tell  $\mathcal{M}$  *which* primitive to compose.  
 617 Neither is sufficient alone — prompts without supporting helpers produced policies that “mention  
 618 rotation” but failed to implement it; helpers without prompt updates often went unused.
- 619 • *Feedback is the only adaptive component*.  $p$ ,  $\mathcal{H}$ , and  $\iota$  are static within a run;  $\phi$  is the only place  
 620 where the researcher gets to change what  $\mathcal{M}$  sees *during* the inner loop, and it uses thresholded  
 621 diagnostics (Listing 4) to do so. This is what mostly drives the run-to-run variance reduction.
- 622 • *The same artifact has different content under each objective*. The Cleanup prompt under  $\Phi_U$  omits  
 623 “rotation” and “maximin” entirely; the same prompt under  $\Phi_{\min}$  devotes its entire opening section

624 to it. The configuration  $c$  is genuinely a function of the welfare objective  $\Phi$ , not a fixed pipeline  
625 parameterized by it — consistent with the information-design framing of Section 3.4.  
626 • *The researcher’s edits are short and targeted.* The full diff between the baseline pipeline and the  
627 best maximin Gemini configuration totals  $\sim 300$  added lines spread over four files;  $\sim 80$  of them  
628 appear in the listings above. The remaining content is API documentation and worked examples,  
629 included for completeness but not strategically novel.

## 630 E Selected Generated Policies

631 The code excerpts below are verbatim outputs of the policy synthesizer  $\mathcal{M}$ , taken from the final  
632 inner-loop iteration of the best-performing run in each condition. To fit the page, we elide repeated  
633 boilerplate (BFS scaffolding, fallback branches) with `...` and add commentary in [brackets];  
634 nothing else has been edited. The listings illustrate the qualitative differences highlighted in Findings  
635 2–4 of Section 4: *static* vs. *rotating* cleaning roles in Cleanup, and the *Voronoi* + *respawn-timer*  
636 structure in Gathering. All four policies were independently generated by the LLM under different  
637 seeds, models, and outer-loop runs.

### 638 E.1 Cleanup, $\Phi_U$ (Gemini): static interleaved roles + dynamic threshold

639 Best run:  $U=3.25$ ,  $E=0.61$ ,  $\min_i R_i = -182$ . The synthesizer assigns *permanent* cleaning roles  
640 via  $\text{agent\_id} \bmod n < \tau$ , where the threshold  $\tau$  depends smoothly on the current waste fraction.  
641 Higher pollution recruits more cleaners, but the *identity* of each cleaner is fixed — agents  $0\text{--}\tau\text{--}1$   
642 always pay the cost. This drives  $U$  to its ceiling but leaves equality at 0.61 and pushes the worst-off  
643 agent to  $\min_i R_i = -182$ .

Listing 5: Cleanup,  $\Phi_U$  – Gemini 3.1 Pro. Static role assignment with waste-tiered threshold (verbatim, abridged).

```

644 1 def policy(env, agent_id) -> int:
645 2     if int(env.agent_timeout[agent_id]) > 0:
646 3         return 7 # STAND while removed
647 4
648 5
649 6     waste_frac = waste_fraction(env)
650 7     n_agents = env.n_agents
651 8
652 9     # [Smooth dynamic threshold: more pollution -> more cleaners]
653 10    if waste_frac > 0.35: threshold_cleaners = n_agents
654 11    elif waste_frac > 0.25: threshold_cleaners = int(n_agents * 0.7)
655 12    elif waste_frac > 0.15: threshold_cleaners = int(n_agents * 0.5)
656 13    elif waste_frac > 0.05: threshold_cleaners = int(n_agents * 0.3)
657 14    else: threshold_cleaners = max(1, int(n_agents * 0.2))
658 15
659 16    # [STATIC role: cleaners are ALWAYS the lowest-id agents -- no rotation]
660 17    is_cleaner = (agent_id % n_agents) < threshold_cleaners
661 18
662 19    if is_cleaner:
663 20        best_orient, n_waste = find_best_clean_orientation(env, agent_id)
664 21        if n_waste >= 1:
665 22            cur_orient = int(env.agent_orient[agent_id])
666 23            if best_orient == cur_orient: return 8 # CLEAN
667 24            if (cur_orient + 1) % 4 == best_orient: return 5 # ROTATE_RIGHT
668 25            elif (cur_orient - 1) % 4 == best_orient: return 4 # ROTATE_LEFT
669 26            else: return 4
670 27            # [Competitor-aware BFS toward waste in own row band]
671 28            c_idx = agent_id % n_agents
672 29            target_r = int(((c_idx + 0.5) / threshold_cleaners) * env.height)
673 30            # ... pick waste cell minimizing dist + |row - target_r| + 50 *
674 31            # closer_cleaners
675 32            # ... bfs_toward and direction_to_action
676 33        else:
677 34            # GATHERER: row-banded sweep, competitor-aware
678 35            # assigned_row = ((g_idx + 0.5) / n_gatherers) * env.height
679 36            # ... pick alive apple minimizing dist + |row - assigned_row| + 50 *
680 37            # closer_gatherers
681 38            ...

```

## 683 E.2 Cleanup, $\Phi_{\min}$ (Gemini): time-rotated, geographically interleaved roles

684 Best run:  $U=3.19$ ,  $E=0.98$ ,  $\min_i R_i=296$ . Switching the outer objective from  $U$  to  $\min_i R_i$  flips  
 685 the role-assignment scheme entirely: `role_idx` now depends on `agent_id + step/50`, so every  
 686 agent rotates through cleaner and gatherer duty in a 50-step cycle. The cleaner indices  $\{1, 4, 8\}$  are  
 687 spatially *interleaved* along the river (rather than contiguous) so that whoever is on duty has a short  
 688 walk to their assigned river slice. This single change accounts for the  $E=0.55 \rightarrow 0.98$  jump and the  
 689  $\min_i R_i=-182 \rightarrow 296$  swing reported in Table 2, with negligible efficiency loss.

Listing 6: Cleanup,  $\Phi_{\min}$  – Gemini 3.1 Pro. Time-rotated roles with interleaved geography (verbatim, abridged).

```

690 1 def policy(env, agent_id) -> int:
691 2     if int(env.agent_timeout[agent_id]) > 0:
692 3         return 7 # STAND while removed
693 4
694 5
695 6     aid = int(agent_id)
696 7     step = int(env._step_count)
697 8     n = int(env.n_agents)
698 9
699 10    # [TIME ROTATION: every 50 steps each agent's role index advances]
700 11    if n == 10:
701 12        role_idx = (aid + step // 50) % 10
702 13        # [Cleaner indices {1,4,8} are SPATIALLY INTERLEAVED, not contiguous,
703 14        # so each on-duty cleaner has a short walk to its river slice]
704 15        if role_idx == 1: role_type, zone_idx = 'C', 0
705 16        elif role_idx == 4: role_type, zone_idx = 'C', 1
706 17        elif role_idx == 8: role_type, zone_idx = 'C', 2
707 18        else:
708 19            role_type = 'G'
709 20            g_map = {0:0, 2:1, 3:2, 5:3, 6:4, 7:5, 9:6}
710 21            zone_idx = g_map[role_idx]
711 22            num_cleaners, num_gatherers = 3, 7
712 23
713 24    if role_type == 'C':
714 25        # [Each on-duty cleaner owns one horizontal slice of the river]
715 26        row_min = int( zone_idx * env.height / num_cleaners)
716 27        row_max = int((zone_idx + 1) * env.height / num_cleaners) - 1
717 28
718 29        # [Score every (row, col, orientation) in the slice by beam yield;
719 30        # fire if facing >=2 waste, else move to a >=2-yield position]
720 31        best_positions, good_positions = set(), set()
721 32        for r in range(row_min, row_max + 1):
722 33            for c in range(10):
723 34                if env.walls[r, c] or env.waste[r, c]: continue
724 35                y_max = max(get_clean_yield(env, r, c, o) for o in range(4))
725 36                if y_max >= 2: best_positions.add((r, c))
726 37                if y_max >= 1: good_positions.add((r, c))
727 38            # ... rotate / CLEAN / bfs_to_target_set as appropriate
728 39        else:
729 40            # GATHERER: collect apples ONLY inside the rotating row band
730 41            # row_min/row_max for num_gatherers slices ...
731 42            ...

```

## 733 E.3 Cleanup, $\Phi_{\min}$ (Sonnet): independent rediscovery of duty rotation

734 Best run:  $U=2.93$ ,  $E=0.83$ ,  $\min_i R_i=154$ . Run with a different policy LLM (Sonnet 4.6) and on  
 735 a different outer-loop trajectory, the synthesizer arrives at the *same* structural insight as the Gemini  
 736 maximin run: a phase counter (`agent_id + step/50`) mod  $n$  rotates which 2 of the 10 agents clean  
 737 at any given time, and a *separate* 200-step zone counter rotates which 5-row band each collector  
 738 sweeps. The two rotation periods (50 and 200 steps) ensure every agent visits every cleaning slot  
 739 *and* every apple zone within one episode — a structural fairness invariant the researcher discovered  
 740 without any explicit instruction to “rotate.”

Listing 7: Cleanup,  $\Phi_{\min}$  – Sonnet 4.6. Phase-rotated cleaning + zone-rotated collection (verbatim, abridged).

```

741 1 def policy(env, agent_id) -> int:
742 2     if int(env.agent_timeout[agent_id]) > 0:
743 3         return 7 # STAND while removed
744 4
745 5
746 6     n = env.n_agents
747 7     step = env._step_count
748 8     wf = waste_fraction(env)

```

```

749 8
750 9 # --- ROLE ASSIGNMENT: fair 2-cleaner rotation ---
751 10 # [50-step phases x 20 phases / episode -> each agent cleans 4 phases (~20%)]
752 11 phase = step // 50
753 12 cleaner_rank = (agent_id + phase) % n # [cycles 0..9 fairly]
754 13 is_cleaner = cleaner_rank < 2 # [exactly 2 cleaners per phase]
755 14
756 15 # [Emergency override: all agents clean when waste crosses the spawn cliff]
757 16 in_emergency = wf >= 0.40
758 17 if in_emergency: is_cleaner = True
759 18
760 19 # [River split: cleaner_slot 0 -> top half, 1 -> bottom half]
761 20 cleaner_slot = (agent_id % 2) if in_emergency else cleaner_rank
762 21
763 22 if is_cleaner and wf > 0.04:
764 23     # [Scan 4 orientations from current position; if best beam shot
765 24     # covers >=3 waste cells, fire; else step to a strictly better neighbor]
766 25     # ... (best_o, best_cnt) = max over 4 directions
767 26     # ... if best_cnt >= 3: rotate to best_o, then CLEAN
768 27     # ... else: scan 4 adjacent cells x 4 orientations for higher yield
769 28     # ... if no waste in beam range: bfs to waste_set in OWN river half,
770 29     # fall back to whole river
771 30
772 31 # --- COLLECTOR LOGIC: 5-zone rotating apple collection ---
773 32 # [5 zones x 200 steps = 1000 steps -> each agent visits every zone once;
774 33 # 2 agents share each zone at any time, reducing competition 10 -> 2]
775 34 bounds = zone_boundaries(env, 5)
776 35 zone_phase = step // 200
777 36 zone = (agent_id + zone_phase) % 5
778 37 z_start, z_end = bounds[zone], bounds[zone + 1]
779 38
780 39 zone_apples = {(int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
781 40                for i in range(env.n_apples) if env.apple_alive[i]
782 41                and z_start <= int(env._apple_pos[i][0]) < z_end}
783 42 # ... bfs_to_target_set(zone_apples), else bfs_nearest_apple as fallback
784 43 return 7

```

#### 786 E.4 Gathering (Gemini): wall-aware Voronoi + spatiotemporal targeting

787 Best run:  $U=2.47$ ,  $E=0.98$ ,  $\min_i R_i=580$ . In Gathering, where costs are symmetric, no role dif-  
788 ferentiation is needed: the entire optimization is spatial and temporal. The synthesizer (i) parti-  
789 tions cells into BFS-Voronoi territories owned by individual agents, (ii) runs a single centralized  
790 BFS from its own position to score every reachable cell with exact wall-aware distances, and  
791 (iii) for both alive and respawning apples in its territory, computes the *earliest collection time*  
792  $\max(\text{walk\_dist}, \text{respawn\_timer})$  and targets the minimum. When the agent must wait on a  
793 dead apple, it steps onto a *non-spawn* adjacent cell rather than blocking a respawn point. This single  
794 sort key —  $\max(\text{distance}, \text{timer})$  — subsumes both “go to nearest apple” and “camp respawn” as  
795 special cases.

Listing 8: Gathering – Gemini 3.1 Pro. Voronoi territories + spatiotemporal priority targeting (ver-  
batim, abridged).

```

796 1 def policy(env, agent_id) -> int:
797 2     if int(env.agent_timeout[agent_id]) > 0:
798 3         return 7
799 4
800 5
801 6 r, c = int(env.agent_pos[agent_id][0]), int(env.agent_pos[agent_id][1])
802 7 orient = int(env.agent_orient[agent_id])
803 8
804 9 # [Wall-aware Voronoi over the gridworld; spawn_points = all apple cells]
805 10 zones = voronoi_zones(env)
806 11 spawn_points = {(int(p[0]), int(p[1])) for p in env._apple_pos}
807 12
808 13 # [Single centralized BFS: exact distance + first move to every reachable cell]
809 14 distances, first_moves = {}, {}
810 15 q = deque([(r, c, 0, None)])
811 16 visited = {(r, c)}
812 17 while q:
813 18     cr, cc, d, fm = q.popleft()
814 19     distances[(cr, cc)] = d
815 20     first_moves[(cr, cc)] = fm
816 21     for dr, dc in [(-1,0), (1,0), (0,-1), (0,1)]:
817 22         nr, nc = cr + dr, cc + dc
818 23         if 0 <= nr < env.height and 0 <= nc < env.width and not env.walls[nr,
819 24         nc]:
820 25             if (nr, nc) not in visited:

```

```

821 24         visited.add((nr, nc))
822 25         q.append((nr, nc, d + 1, fm if fm is not None else (dr, dc)))
823 26
824 27     # [Both alive AND respawning apples in MY zone, with their respawn timer]
825 28     my_zone_spawns = []
826 29     for i in range(env.n_apples):
827 30         pr, pc = int(env._apple_pos[i][0]), int(env._apple_pos[i][1])
828 31         if zones.get((pr, pc)) == agent_id and (pr, pc) in distances:
829 32             my_zone_spawns.append((pr, pc, int(env.apple_timer[i]), distances[(pr,
830 33                 pc)]))
831 34
832 35     # [SPATIOTEMPORAL PRIORITY: earliest collection time first.
833 36     # score = max(walk_distance, respawn_timer); ties broken by sooner timer]
834 37     my_zone_spawns.sort(key=lambda x: (max(x[3], x[2]), x[2], x[3]))
835 38
836 39     for pr, pc, timer, dist in my_zone_spawns:
837 40         if timer == 0: # [Alive apple: walk straight to it]
838 41             if dist == 0: return 7
839 42             fm = first_moves[(pr, pc)]
840 43             return direction_to_action(fm[0], fm[1], orient)
841 44         else: # [Dead apple: camp on safe adjacent cell]
842 45             # ... pick nearest neighbor (nr,nc) that is NOT in spawn_points,
843 46             # ... navigate there and STAND while waiting for respawn
844 47             # [Global poaching fallback if zone is empty: nearest alive apple anywhere]
845 48             ...
846 49

```

## 848 E.5 Cross-condition observations

849 Several patterns are visible across these four listings (and confirmed by inspecting the remaining 8  
850 runs not shown):

- 851 • *Role assignment encodes the welfare objective.* Static  $\text{agent\_id} < \tau$  (Listing 5) maximizes  
852  $U$  but harms  $\min_i R_i$ . Time-rotated  $(\text{agent\_id} + \text{step} // T) \% n$  (Listings 6, 7) maximizes  
853  $\min_i R_i$  at  $\leq 1\%$  efficiency cost (Gemini).
- 854 • *Convergent rediscovery across LLMs.* Gemini and Sonnet, on independent runs, both arrive at  
855 the  $(\text{id} + \text{phase}) \% n$  duty-rotation idiom with similar phase lengths ( $T \approx 50$  steps) under  
856 maximin — and both omit it under efficiency.
- 857 • *Game structure dictates strategy class.* Cleanup policies are dominated by role-assignment logic  
858 (cleaner vs. gatherer); Gathering policies are dominated by territory partitioning and respawn-  
859 timer reasoning, with no role differentiation at all. The researcher does not need to be told which  
860 class of strategy to write.
- 861 • *Earliest-collection-time as a unifying score.* The Gathering policy’s  $\max(\text{walk}, \text{timer})$  key (List-  
862 ing 8) is structurally identical across the 4 Gathering runs (both Gemini and Sonnet), differing  
863 only in how the Voronoi diagram is computed — Manhattan approximation, fully BFS-based, or  
864 cached helper.